THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE CHAPTER TEST SET A

Chapter 6: **Pressure, Winds, Storms and Cyclones** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A force of 100 N acts perpendicularly on an area of 2 m². The pressure is: 1

(a) 200 N/m² **(b)** 50 N/m² **(c)** 25 N/m² **(d)** 100 N/m²

2. Pressure is defined as: 1

(a) force times area **(b)** area per unit force
(c) mass per volume **(d)** force per unit area

3. Liquids exert pressure: 1

(a) in all directions **(b)** sideways only
(c) downwards only **(d)** upwards only

4. The envelope of air around Earth is the: 1

(a) hydrosphere **(b)** atmosphere **(c)** biosphere **(d)** lithosphere

5. 1 millibar equals: 1

(a) 1000 Pa **(b)** 100 Pa **(c)** 10 Pa **(d)** 1 Pa

6. A sea breeze blows: 1

(a) from land to sea during the day **(b)** from sea to land at night
(c) from sea to land during the day **(d)** only in winter

7. During a storm it is safer to: 1

(a) go under a tall tree
(b) keep doors and windows open (reducing pressure difference)
(c) stand on the roof
(d) seal everything tightly

8. Upper parts of thunderclouds carry mostly: 1

(a) no charges **(b)** only neutral dust
(c) positive charges **(d)** negative charges

9. A cyclone forms over: 1

(a) polar ice **(b)** warm ocean waters
(c) mountain tops **(d)** cold deserts

10. The cyclone Amphan of 2020 had peak winds of about: 1

(a) 70 km/h **(b)** 270 km/h **(c)** 27 km/h **(d)** 500 km/h

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. Define pressure and write its formula. 2

12. State the SI unit of pressure and its relation to newton and metre. 2

13. Why do broad straps on a school bag help the shoulders? 2

14. Why do liquids exert pressure on the sides of a container? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. A force of 300 N acts on an area of 1.5 m². Find the pressure. If the area is doubled, what is the new pressure? 3

16. Why is it easy to drive a nail by its pointed end? Explain with pressure. 3

17. Explain with reasons why the base of a dam is thicker. 3

OR

Explain why a person on a lower floor gets a stronger stream of water than one on a higher floor. 3

18. Calculate the force exerted by the atmosphere on 15 cm × 15 cm area, as given in the book, and state its equal mass. 3

OR

Describe the activity showing atmospheric pressure with a sucker or a balloon. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. Define pressure. Explain how force and area affect pressure with two examples each. 5

OR

Explain liquid pressure. Write its properties and its applications in dams and water tanks. 5

20. What is atmospheric pressure? Describe two activities that show it. 5

OR

Explain the formation of winds and describe sea and land breezes with a diagram. 5

END OF QUESTION PAPER — Total: 40 Marks

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